A counterexample to the $1/(e2^n)$ mixed-integer Grünbaum bound

Working note for Phillip Kerger and Hongyu Cheng 7 November 2026

For a bounded Borel set $D \subset \mathbb{R}^n \times \mathbb{R}^d$, write

$$\mu_{\mathrm{MI}}(D) = \sum_{z \in \mathbb{Z}^n} \mathrm{vol}_d\{x : (z,x) \in D\}, \qquad \mathrm{depth}_C(y) = \inf_{\substack{H \text{ closed halfspace} \\ y \in H}} \frac{\mu_{\mathrm{MI}}(C \cap H)}{\mu_{\mathrm{MI}}(C)}.$$

A halfspace is proper: its defining linear functional is nonzero. All volumes below are ordinary Lebesgue volumes in the indicated continuous coordinates.

Theorem 1. *There is a full-dimensional rational polytope $C \subset \mathbb{R}^{76} \times \mathbb{R}^d$, where $d = 1{,}372{,}131 + 2^{93}$, such that $S = C \cap (\mathbb{Z}^{76} \times \mathbb{R}^d)$ is nonempty, every nonempty integer fiber has nonempty d-dimensional interior, and*

$$\sup_{y \in S} \mathrm{depth}_C(y) < \frac{1}{e2^{76}}.$$

Consequently the specific $1/e$ bound and the larger sharp $\Gamma_d = (d/(d+1))^d$ bound both fail.

This does not disprove a bound $c\,2^{-n}$ for some smaller positive absolute constant c. The construction and strict mixed-integer-volume conclusion have been verified in Lean; the precise scope is recorded on page 6.

The idea of the construction

There will be just one exceptional integer fiber, over the origin. Every other integer base will be exposed: a linear functional on the integer coordinates uniquely maximizes there. A sufficiently large multiple of that functional allows us to isolate a continuous cap in its fiber. We therefore have two separate tasks: make every point over the origin shallow, and provide suitable caps at all the exposed bases.

Before a final simplex factor is added, the intended fibers have these volumes:

Integer bases	Number	Volume of each fiber
Origin	1	b
Nonzero vertices of the unit cube	$2^{76} - 1$	bm
Negative coordinate units	76	$b\alpha$

The cube matches the factor 2^{76} in the proposed bound. The additional negative-unit fibers create a surplus in the total mass:

$$\mu_{\mathrm{MI}}(C_0) = bW, \qquad W = 2^{76}m + \delta, \qquad \delta = 1 + 76\alpha - m.$$

We shall construct four halfspaces that cover the entire origin fiber. Each contains a fraction $27/64$ of that fiber, exactly 19 outer fibers, and no other fibers. Its mass is consequently bL, where $L = 27/64 + 19\alpha$. The quantitative goals are

$$0 < \alpha \le m, \qquad \delta > 0, \qquad L < m/e.$$

The first two make each exposed fiber smaller than 2^{-76} of the total; the third deals with the origin. A common ℓ-simplex factor then supplies caps of fraction $\Gamma_\ell \to 1/e$ at the exposed bases, without changing the origin-cut ratios. For sufficiently large finite ℓ, the uniform bound is

$$\frac{\max\{L, m\Gamma_\ell\}}{2^{76}m + \delta} < \frac{1}{e2^{76}}.$$

The proof below realizes this scheme. The main compatibility issue is that taking a convex hull must not enlarge the origin fiber. A grouped Minkowski containment will resolve exactly this issue. We keep the parameters symbolic until their geometric roles are clear, and give exact rational values in Section 4.

1. Four source types and the two continuous factors

Let $B_0 \subset \mathbb{R}^3$ be the standard tetrahedron translated to have centroid zero, with barycentric coordinates $\lambda_0, \dots, \lambda_3$. Thus $f_j = \lambda_j - 1/4$ is linear and $\sum_j f_j = 0$. For a small rational $\varepsilon > 0$ and $a \in \{0,1,2,3\}$ define

$$T_a = \{x \in B_0 : \lambda_j(x) \le 1/4 \ (j \ne a)\}, \qquad \lambda_j(c_a) = \begin{cases} 1/4 + 3\varepsilon & j = a, \\ 1/4 - \varepsilon & j \ne a. \end{cases}$$

The three coordinates with $j \ne a$ identify T_a with $[0,1/4]^3$ and B_0 with the standard simplex in those coordinates. Hence

$$\theta := \frac{|T_a|}{|B_0|} = \frac{(1/4)^3}{1/6} = \frac{3}{32}, \qquad \frac{|B_0 \cap \{f_j \ge 0\}|}{|B_0|} = \left(\frac{3}{4}\right)^3 = \frac{27}{64}. \tag{1}$$

The second identity follows by homothety about vertex j. The reason for these sources is that T_a lies in $\{f_j \le 0\}$ for all $j \ne a$: a source of type a can be excluded by three of the four cuts. Its anchor c_a lies just inside these three planes. Contraction about this anchor will give strict separation.

Choose rational parameters and an integer $h \ge 1$ satisfying

$$\begin{aligned} &0 < \varepsilon < 1/4, \quad 1/2 < s < 1, \quad 1/2 < q < 1, \quad u = 2(1-q), \\ &0 < t < \eta < 2(1-s), \qquad t(1/4 + 3\varepsilon) \le 2(1-s)(1/4 - \varepsilon). \end{aligned} \tag{2}$$

The box $F = [-1,1]^h$ will tune volumes; its coordinates never enter the four origin cuts. Define

$$\begin{aligned} B &= B_0 \times F, & P_a &= [c_a + s(T_a - c_a)] \times qF, \\ A_a &= t(T_a - c_a) \times uF, & K(a,k) &= 2P_a + (k-1)(c_a, 0) \quad (k \ge 1). \end{aligned} \tag{3}$$

All these sets are rational polytopes with nonempty interior; c_a is an interior point of T_a. Writing $b = |B| > 0$, scaling and product volumes give

$$|K(a,k)| = bm, \quad |A_a| = b\alpha, \qquad m = (2s)^3\theta(2q)^h, \quad \alpha = t^3\theta u^h. \tag{4}$$

Since $2q > 1$ and $u < 1$, increasing h increases m and decreases α. The relation $u = 2(1-q)$ keeps these changes compatible with convexification. The later simplex factor has a different job: it supplies small containing caps at exposed bases, multiplying every existing volume by the same factor.

2. The compatibility needed at the origin

Lemma 1 (Grouped containment)**.** *For $k \ge 1$ and arbitrary source types $a, a_1, \dots, a_k$,*

$$K(a,k) + \sum_{r=1}^{k} A_{a_r} \subset (k+1)B. \tag{5}$$

Proof. Set $c_- = 1/4 - \varepsilon$ and $c_+ = 1/4 + 3\varepsilon$. The final assumption in (2) is

$$tc_+ \le 2(1-s)c_-. \tag{6}$$

For $p \in c_a + s(T_a - c_a)$ and $v_r \in t(T_{a_r} - c_{a_r})$ we have $\lambda_j(p) \ge (1-s)\lambda_j(c_a)$ and $f_j(v_r) \ge -t\lambda_j(c_{a_r})$. Therefore, for $y = (2p + (k-1)c_a + \sum_r v_r)/(k+1)$,

$$\begin{aligned} (k+1)\lambda_j(y) &\ge (k+1-2s)c_- - ktc_+ \\ &\ge \big[k+1-2s-2k(1-s)\big]c_- = (k-1)(2s-1)c_- \ge 0. \end{aligned}$$

Thus $y \in B_0$. The free-coordinate radius is at most $2q + ku \le k+1$, since $k+1-2q-ku = (k-1)(2q-1) \ge 0$. These two statements prove (5). □

The convex body and its actual fibers

Index the integer coordinates by $I = \{0,1,2,3\} \times \{1,\dots,19\}$; let $\tau(i)$ be the first coordinate. For each nonempty $E \subset I$, fix once and for all one member, let $a(E)$ be that member's type, and set $z_E = \sum_{i\in E} e_i$. The choice $a(E) \in \tau(E)$ ensures that if E has no coordinate of type j, then $a(E) \neq j$; cut j will therefore exclude its source. The translation $(|E|-1)c_{a(E)}$ in (3) makes the corresponding negative continuous score grow with $|E|$. Define $C_0 \subset \mathbb{R}^{76} \times \mathbb{R}^{3+h}$ to be the convex hull of

$$\{0\} \times B, \qquad \{z_E\} \times K(a(E),|E|) \quad (\varnothing \neq E \subset I), \qquad \{-e_i\} \times A_{\tau(i)} \quad (i \in I). \tag{7}$$

Why is (5) the relevant condition? The projected generators satisfy

$$0 = \frac{z_E + \sum_{i\in E}(-e_i)}{|E|+1}.$$

Such groups can contribute to the origin fiber when we take the convex hull. Lemma 1 says their continuous averages remain inside B. The next argument shows that every contribution to the origin can be regrouped in precisely this way.

Lemma 2 (Exact integer fibers)**.** *The only integer bases of C_0 are 0, the nonzero cube vertices z_E, and the negative units $-e_i$. Their fibers are exactly the sets in (7). The polytope C_0 has nonempty ambient interior.*

Proof. Every coordinate of its projection lies in $[-1,1]$. If an integer projected point has a negative coordinate, that coordinate is -1, which forces every positive-weight generator in a convex representation to be the unique negative unit with that coordinate. Otherwise the point is a cube vertex. For nonempty E, the functional with coefficients 1 on E and $-1/2$ outside E uniquely exposes z_E among all bases in (7): its value there is $|E| \geq 1$, whereas negative units have value at most $1/2$. The functional $-z_i$ uniquely exposes $-e_i$. Their fibers are therefore exact.

At the origin, aggregate a convex representation into weights w_0, w_E, v_i and points $x_0 \in B$, $x_E \in K(a(E),|E|)$, $y_i \in A_{\tau(i)}$. Convexity of the assigned fibers permits this aggregation. Zero integer coordinate and total weight one imply

$$v_i = \sum_{E\ni i} w_E, \qquad w_0 + \sum_E (|E|+1)w_E = 1.$$

The continuous point becomes

$$w_0 x_0 + \sum_E w_E \left(x_E + \sum_{i\in E} y_i \right) \in B$$

by Lemma 1; unused zero-weight points may be chosen arbitrarily. The reverse inclusion follows from (7). Finally the origin fiber spans all continuous directions, and the projected bases span all integer directions. Thus the affine hull of C_0 is the whole ambient space. A full-dimensional convex polytope has nonempty interior. □

3. Four small cuts cover the origin fiber

Let $R = 231$. For each $j \in \{0,1,2,3\}$ define the proper closed halfspace

$$H_j = \left\{ (z,x,w) : f_j(x) + \sum_{i\in I} \beta_i z_i \geq 0 \right\}, \qquad \beta_i = \begin{cases} -R & \tau(i) = j, \\ \eta\varepsilon & \tau(i) \neq j. \end{cases} \tag{8}$$

Here $x \in \mathbb{R}^3$ and $w \in \mathbb{R}^h$. Call a coordinate with $\tau(i) = j$ *paid*: its outer fiber will be included by this cut. The continuous functional f_j is nonzero. Our assumptions give

$$0 < t < \eta < 2(1-s) < 1, \qquad \eta\varepsilon < 1. \tag{9}$$

For $j \neq a$, the definitions give

$$\begin{aligned} f_j(P_a) &\le -(1-s)\varepsilon, \qquad f_j(A_a) \le t\varepsilon, \\ f_j(K(a,k)) &\le -(k+1-2s)\varepsilon. \end{aligned} \tag{10}$$

These inequalities mean pointwise upper bounds on the indicated sets, ignoring their free coordinates. For every a, j, also $|f_j(A_a)| \le 2t$ and $f_j(K(a,k)) \le k+1$: the latter follows from $P_a \subset B_0 \times \mathbb{R}^h$, $c_a \in B_0$, and $|f_j| \le 1$ on B_0. There are two ways to exclude a cube fiber: a paid coordinate contributes a large negative integer term, or its source contributes a negative continuous term. Using the exact census, the complete accounting is:

- At an unpaid negative unit, the score in (8) is at most $(t-\eta)\varepsilon < 0$. At a paid negative unit it is at least $R - 2t > 0$, so precisely the 19 paid outer fibers are included completely.
- If E contains a paid coordinate and $k = |E|$, its entire fiber has score at most $(k+1) - R + (k-1)\eta\varepsilon \le 2k - R < 0$, since $k \le 76$.
- If E contains no paid coordinate, then $a(E) \neq j$, so its score is at most $[k\eta - (k+1-2s)]\varepsilon$. By (9) this is strictly less than $-(k-1)(2s-1)\varepsilon \le 0$, including when $k = 1$.

Thus every nonzero cube fiber is strictly excluded. At the origin, (8) reduces to $f_j \ge 0$. Consequently (1) and (4) give

$$\mu_{\mathrm{MI}}(C_0 \cap H_j) = bL, \qquad L := \frac{27}{64} + 19\alpha. \tag{11}$$

Since $\sum_j f_j = 0$, every point of the origin fiber belongs to at least one of these four cuts. All halfspaces are closed; no boundary mass has been removed in this calculation.

4. Choosing the parameters to balance the masses

We now specify the rational values promised in (2):

$$\begin{gathered} h = 1{,}372{,}128, \quad \varepsilon = 10^{-10}, \quad s = \frac{5000001}{10^7}, \quad t = \frac{9999997}{10^7}, \\ \eta = \frac{3999999}{4000000}, \quad q = \frac{500001}{10^6}, \quad u = 2(1-q). \end{gathered}$$

The elementary inequalities in (2) hold. For the less immediate one,

$$2(1-s)c_- - tc_+ = \frac{2460000011}{10^{17}} > 0.$$

Recall the mass surplus and total, now justified by the exact fiber census:

$$\delta = 1 + 76\alpha - m, \qquad W = 2^{76}m + \delta, \qquad \mu_{\mathrm{MI}}(C_0) = bW. \tag{12}$$

The exact power certificates give the following strict rational intervals:

$$\begin{aligned} m_- := \frac{1458094201550}{10^{12}} < m < \frac{1458094201551}{10^{12}} =: m_+, \\ \alpha_- := \frac{6027739233}{10^{12}} < \alpha < \frac{6027739234}{10^{12}} =: \alpha_+. \end{aligned} \tag{13}$$

Page 6 supplies a short exact-arithmetic reproduction and its enclosure invariant. In particular $0 < \alpha < m$. Put $U = 27182819/10^7$. The exponential series gives

$$e < \sum_{j=0}^{10} \frac{1}{j!} + \frac{12}{11 \cdot 11!} < U;$$

the tail starting with $1/11!$ is bounded by a geometric series of ratio $1/12$.

The margins required below are simple rational subtractions:

$$\begin{aligned} \frac{m_-}{U} - \frac{27}{64} - 19\alpha_+ &= \frac{11151455803863}{13591409500000000000} > 0, \\ 1 + 76\alpha_- - m_+ - \frac{m_+}{2^{17}} &= \frac{374312936753}{131072000000000000} > 0. \end{aligned} \tag{14}$$

Combining (13)–(14) with $e < U$ proves

$$L < \frac{m}{e}, \qquad \delta > \frac{m}{2^{17}} > 0. \tag{15}$$

5. Why a finite simplex factor completes the example

The surplus δ has a specific purpose: it lets us use a simplex cap slightly larger than $1/e$. The required comparison is $e\Gamma_\ell < 1 + \delta/(2^{76}m)$. The estimate $\delta/m > 2^{-17}$ suggests $\ell = 2^{76+17}$; this explains the large exponent rather than treating it as an unexplained parameter. Let $J \subset \mathbb{R}^\ell$ be the standard simplex, with $\ell = 2^{93}$, and set $C = C_0 \times J$, grouping all continuous coordinates together. Its actual integer fibers are precisely the fibers of Lemma 2 times J. It is a full-dimensional rational polytope and all its nonempty fibers have full dimension. Its mass is $bW|J| > 0$.

Lemma 3 (Isolating a simplex cap)**.** *At every mixed-integer point of C over a nonzero base, a proper closed ambient halfspace contains the point and has mass at most $bm\Gamma_\ell|J|$, where $\Gamma_\ell = (\ell/(\ell+1))^\ell$.*

Proof. Let z_* be its base and $v \in J$ its simplex coordinate. Some barycentric coordinate $\rho_j(v)$ is at least $a = 1/(\ell+1)$. The cap $\{v' \in J : \rho_j(v') \geq a\}$ is the homothetic image $av_j + (1-a)J$, where v_j is the corresponding vertex. Its volume is $\Gamma_\ell|J|$.

By Lemma 2, a linear functional g on integer coordinates strictly exposes z_*. There are only finitely many other bases. Thus $\Delta = \min_{z \neq z_*}[g(z_*) - g(z)] > 0$. Choose $M > 1/\Delta$. The halfspace

$$\rho_j(v') - a + M[g(z') - g(z_*)] \geq 0 \tag{16}$$

is proper, contains the given point, cuts out exactly that cap at z_*, and misses every other fiber, since $\rho_j(v') \leq 1$ on J. Its affine expression is equivalent to a linear functional with a real threshold; the continuous linear part of ρ_j is nonzero. The old fiber at z_* has volume either bm or $b\alpha \leq bm$. Multiplication of product volumes proves the claim. □

For every positive integer k,

$$e\left(\frac{k}{k+1}\right)^k < 1 + \frac{1}{k}. \tag{17}$$

Indeed $\log(1 + 1/k) > 1/(k+1)$, obtained by integrating $1/x$ on $[1, 1+1/k]$. Thus $(k+1)\log(1+1/k) > 1$, which is equivalent to (17). By (15) and $\ell = 2^{76+17}$,

$$e\Gamma_\ell < 1 + \frac{1}{\ell} < 1 + \frac{\delta}{2^{76}m} = \frac{W}{2^{76}m}. \tag{18}$$

Proof of Theorem 1. At every origin-fiber point, extend a containing cut (8) without using the J coordinates; (11) gives normalized mass L/W. All other points are covered by Lemma 3, with normalized mass at most $m\Gamma_\ell/W$. By (15), $W > 2^{76}m$, and (18), their uniform upper bound is

$$r := \frac{\max\{L, m\Gamma_\ell\}}{W} < \frac{1}{e2^{76}}. \tag{19}$$

Therefore $\sup_{y \in S} \operatorname{depth}_C(y) \leq r$. There is no limiting construction or assumption that a deepest point exists. The origin belongs to S. All fibers have full dimension, so $\dim_H S = d$. Euclidean $\mathcal{H}_d$, normalized to agree with Lebesgue measure on d-dimensional affine planes, is exactly the mixed-integer measure on this finite union of disjoint fibers. Finally $\Gamma_d > 1/e$, since $d\log(1+1/d) < 1$. This also refutes the sharp bound. □

Appendix A. Reproducing the finite arithmetic

The large powers in (4) require no floating-point computation. Set $Q = 10^{30}$. For nonnegative rational intervals, multiply their endpoints and round the lower endpoint down and the upper endpoint up to multiples of $1/Q$. Positivity preserves the enclosure at every multiplication; induction through binary exponentiation therefore gives a rigorous enclosure of each power.

The following standalone Python code (standard library only) verifies (13) and the two margins. Every assertion is an exact rational comparison. The Lean certificate independently proves its individual rational inequalities inside the kernel; this Python program is not part of Lean's trusted base.

```
from fractions import Fraction as F
from math import floor, ceil, factorial
Q = 10**30

def mul(A, B):
    return (F(floor(Q*A[0]*B[0]), Q),
            F(ceil(Q*A[1]*B[1]), Q))

def power(x, n):
    out, base = (F(1), F(1)), (x, x)
    while n:
        if n % 2: out = mul(out, base)
        n //= 2
        if n: base = mul(base, base)
    return out

s, t, q = F(5000001,10**7), F(9999997,10**7), F(500001,10**6)
h, theta = 1372128, F(3,32)
G, D = power(2*q,h), power(2*(1-q),h)
M = tuple((2*s)**3*theta*x for x in G)
A = tuple(t**3*theta*x for x in D)
ml, mh = F(1458094201550,10**12), F(1458094201551,10**12)
al, ah = F(6027739233,10**12), F(6027739234,10**12)
assert ml < M[0] <= M[1] < mh
assert al < A[0] <= A[1] < ah
U = F(27182819,10**7)
assert sum((F(1,factorial(j)) for j in range(11)), F(0)) \
       + F(12,11*factorial(11)) < U
assert ml/U - F(27,64) - 19*ah == \
       F(11151455803863,13591409500000000000)
assert 1 + 76*al - mh - mh/2**17 == \
       F(374312936753,131072000000000000)
```

Appendix B. Formal verification and scope

The final unconditional Lean theorem is

`SharpOC.BoxCounterexample.one_over_e_counterexample`

in `lean/SharpOC/BoxMeasure.lean`. It proves the stated dimensions, an actual compact convex body with nonempty ambient interior, positive finite mass summed over *all* integer bases, full-dimensional nonempty fibers, and one uniform strict bound below $1/(e2^{76})$ using proper closed ambient halfspaces at every actual mixed-integer point.

The pinned project uses Lean 4.34.1 and mathlib commit `d13f23b723b8`. Both `lake build` and `lake env lean CheckAxioms.lean` passed. The final theorem depends only on `propext`, `Classical.choice`, and `Quot.sound`; there are no admitted proofs or extra axioms. The formal theorem uses canonical Lebesgue measure in continuous product coordinates. Rational polyhedrality and the Euclidean Hausdorff-normalization identification are established in the prose proof, rather than asserted as additional Lean conclusions.

The formalization is commit `d27b977`, PR #7 in Hongyu Cheng's repository. The analytic result is recorded as fact `a5291bca09ba4b1e`; the formal proof map is `lean/ONE_OVER_E_AUDIT.md`.